LucyTrace 🐾

Turn "Ruff Drafts" into the "Director's Cut"



LucyTrace is a high-performance pipeline for converting raster images (PNG, JPG, WEBP, etc.) into clean vector SVGs and print-ready coloring pages.

Named after our Director of Wellness (Lucy the Cavachon), this tool ensures your images are well-groomed, free of noise, and ready for:

1. **Laser Cutting/Engraving:** Compatible with LightBurn, XTool Creative Space, and Glowforge.
2. **CAD/CNC:** Optimized node counts for smoother import into **Autodesk Fusion** sketches.
3. **Print:** Generates high-resolution, centered PDFs and PNGs for coloring books.

🚀 Quick Start

1. Automated Setup (First Time)

Run the cross-platform setup wizard:

```
# Windows
python setup.py

# macOS / Linux
python3 setup.py
```

The wizard will:

- Check for ImageMagick, Potrace, and Inkscape.
- Prompt to create a virtual environment with OS-friendly defaults (`venv_win` , `venv_mac`).
- Install dependencies from `requirements.txt` .
- Optionally generate a one-click Desktop shortcut with the LucyTrace icon.

2. Launching LucyTrace

You can run LucyTrace using:

- **Desktop App (CustomTkinter):** Double-click your Desktop shortcut or run `python lucytrace.py`
- **Command Line / Batch Script:** Double-click `lucytrace.bat` / `lucytrace.sh` or run `python lucytrace.py --cli`

3. Collect Output

- `output_svg/` -> Vector files (Optimized for Laser/CAD).
- `output_png/` -> High-res raster files.
- `output_pdf/` -> Print-ready documents.

⚙️ The Two Profiles

LucyTrace uses a unified engine but offers two distinct "grooming" styles:

1. Base Profile ("The Fast Track")

- **Best For:** High-quality scans, crisp digital line art, black ink on white paper.
- **Logic:** Applies a strict brightness threshold (default 60%). Pixels darker than the limit become black; everything else becomes white.
- **Pros:** Extremely fast and predictable on clean input.

2. Advanced Profile ("The Director's Cut")

- **Best For:** Photos of drawings (bad lighting), "muddy" scans, comic book panels, or inverted images (white lines on black).
- **Logic:**
  - **Auto-Levels:** Stretches contrast to fix lighting issues (makes dark grays black, light grays white).
  - **Smart Detection:** Automatically detects if an image is Color or B&W.

- **Posterization:** If color is detected, it flattens gradients into chunks ("stained glass" style) before tracing to prevent noise.

🛠️ Configuration (`lucytrace.ini`)

You don't need to edit the Python code to change settings. All major toggles are in `lucytrace.ini` :

```
[General]
profile = adv           ; Default profile if running via CLI
overwrite = false       ; Set to true to force reprocessing existing files
export_width = 3000     ; Width of the final PNG (3000px is approx 10 inches at 300 DPI)

[ImageProcessing]
threshold_percent = 60% ; The cutoff for black vs white

[Modes]
invert = off            ; Set to 'on' if your input is white lines on a black background
mode = auto             ; Can force 'color' or 'bw' manually here
```

📂 Folder Structure

The project uses the following structure (folders are auto-created if missing):

```
<project folder>/
├── input_images/   # Source images (auto-moved here from root)
├── cleaned_png/    # Intermediate black/white PNGs (useful for debugging)
├── output_svg/     # Traced vector output files (Optimized for CAD/Laser)
├── output_png/     # Exported PNGs for printing
├── output_pdf/     # Exported PDFs for printing
│
├── lucytrace.py    # The Master Script
├── lucytrace.ini   # Configuration File
├── lucytrace.bat   # Windows Launcher
├── lucytrace.sh    # Mac/Linux Launcher
└── README.md
```

📦 Requirements

LucyTrace requires **Python 3**, **ImageMagick**, **Potrace**, and **Inkscape**.
All tools must be installed and accessible via your system PATH.

Installation Guide

🪟 Windows Setup

1. **Python 3:** Download from python.org. Ensure **"Add to PATH"** is checked.
2. **ImageMagick:** Download the DLL installer from imagemagick.org. Check **"Install legacy utilities"** and **"Add to PATH"**.
3. **Inkscape:** Install from inkscape.org.
4. **Potrace:**
   - Download the Windows zip from SourceForge.
   - Extract it (e.g., to `C:\Program Files\potrace`).
   - Add that folder to your System Environment **PATH**.

🍎 macOS Setup

1. **Homebrew:** If you don't have it, install it first.
2. **Install Tools:**

   ```
   brew install python imagemagick potrace --cask inkscape
   ```

3. **Link Inkscape:** macOS needs this link to find the tool:

   ```
   sudo ln -s /Applications/Inkscape.app/Contents/MacOS/inkscape /usr/local/bin/inkscape
   ```

4. **Permissions:** Make the launcher executable:

   ```
   chmod +x lucytrace.sh
   ```

🐧 Linux Setup (Debian/Ubuntu)

```
sudo apt update
sudo apt install python3 python3-pip imagemagick potrace inkscape
```

Support the Maker (and Lucy!)

I develop these tools to improve my own parametric workflows and love sharing them with the community. If you find LiveUtilities useful and want to say thanks, feel free to **buy Lucy a dog treat on Ko-fi**! This is completely optional and supports my Chief Wellness Officer in maintaining mandatory play breaks. Your appreciation and feedback are more than enough.

Happy Making!
— EdJ